BITS Pilani – Work Integrated Learning Programmes

M.Tech in Artificial Intelligence & Machine Learning

Course: Machine Learning Operations (MLOps) - S1-26_AIMLZG523
Assignment 2: EduAssist AI: Systematic Evaluation, Token Economics, and Production Promotion of Academic LLM Prompts

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Target Model: Gemini 3.5 Flash Lite (gemini-3.5-flash-lite)

Tracking Server: MLflow Tracking & Prompt Registry (eduassist_llmops_assignment)

Repository: https://github.com/krishnasai199811/EduAssist-AI-Assignment-2

1. Executive Summary & Architecture

In production LLMOps architectures, prompt design directly dictates inferencing latency, serving infrastructure costs, and downstream pipeline determinism. This empirical investigation evaluates three successive prompt iterations (v1_Basic, v2_Structured, and v3_JSON) for the EduAssist tutoring microservice. The system decouples operational logic from configuration by externalizing pricing tiers, schema constraints, and tracking parameters into config.json and academic_data.json. Dynamic environment variable injection guarantees secure authentication without exposed secrets. All prompt executions were systematically tracked via an MLflow experiment to record high-resolution latency, token distributions, request-level financial expenditures in Indian Rupees (INR), and schema parsing validity across a standard undergraduate computer science dataset.

2. Task 1: Prompt Economics & Latency Analysis

Empirical baseline testing utilizing Prompt v1 reveals an acute cost asymmetry between ingestion and completion tokens. Billed at standard pricing tiers of ₹6.25 per million input tokens and ₹25.00 per million output tokens:

- **Cost Asymmetry:** Across five queries, baseline input processing accounted for only 145 tokens (mean: 29.0 tokens/query), totaling ₹0.000906 or **1.12%** of aggregate expenditure. Conversely, unconstrained model generation consumed 3,187 tokens (mean: 637.4 tokens/query), accumulating ₹0.079675, which constitutes **98.88%** of total spend.
- **Latency Correlation:** Wall-clock response latency exhibits a strong linear correlation with generated output volume rather than prompt length. Mean baseline latency was 2.7677 seconds (median: 2.9938s, standard deviation: 0.8437s). The fastest

query (Q1) generated 166 tokens in 1.4615s, whereas the slowest query (Q5: ETL Stream vs. Batch) generated 869 output tokens in 3.5633s.

Because auto-regressive decoding generates tokens sequentially, generation verbosity acts as the primary latency bottleneck. Economically, this 98.88% output dominance demonstrates that adding structural constraints to input prompts is virtually negligible in cost, whereas unconstrained completions drive exponential financial waste.

3. Task 2: Iterative Prompt Engineering Progression

To curb output inflation and improve pedagogical clarity, two architectural revisions were evaluated against the baseline:

Metric	Prompt v1 (Basic)	Prompt v2 (Structured)	Prompt v3 (JSON Constrained)
Prompt Template Focus	Open-ended question answering	Persona, undergraduate clarity	Strict JSON schema, confidence
Mean Input Tokens	29.0	53.0	91.0
Mean Output Tokens	637.4	366.0	166.2
Mean Latency (s)	2.7677	1.8957	1.3820
Mean Cost / Query (INR)	₹0.016116	₹0.009481	₹0.004724
Valid JSON Extraction	0.0%	0.0%	100.0%

- **v1 -> v2:** Adding an academic persona and reading level constraints expanded input prompt size to 53.0 tokens, yet pruned rambling output by 42.58% (366.0 tokens), accelerating response latency by 31.51% (1.8957s).
- **v2 -> v3:** Enforcing a rigorous schema (answer, confidence, key_points) raised input length to 91.0 tokens. However, it eliminated conversational fluff, reducing output

length by 54.59% over v2 and **73.93% over v1** (166.2 tokens). Latency dropped to 1.3820s (**50.07% faster than baseline**).

The ₹0.000388 incurred from adding 62 input tokens saved ₹0.011780 in output fees, yielding a net savings of ₹0.011392 per invocation (**70.69% total cost reduction**).

4. Task 3: Production Promotion Justification (alias: production)

Prompt v3 was formally assigned the production alias within MLflow based on four critical criteria:

1. **API Reliability:** While v1 and v2 yielded free-form Markdown incompatible with automated ingestion (0% JSON compliance), Prompt v3 achieved a **100.0% valid JSON parse rate**, eliminating parsing exceptions across downstream services.
2. **SLA Compliance:** Latency was cut by 50.07%, consistently meeting interactive UI thresholds.
3. **Enterprise Unit Economics:** At 10,000 monthly queries, baseline Prompt v1 incurs ₹161.16, whereas Prompt v3 costs just **₹47.24**—securing a recurring **70.69% monthly cost reduction (₹113.92 saved per 10k queries)**.
4. **Automated Triage via Metadata:** The programmatic confidence rating (high | medium | low) enables automated triage, routing uncertain responses to human teaching assistants.

Prompt v3 represents the pareto-optimal balance of schema determinism, low latency, and operational cost efficiency.